Rushabh Shah

(617)764-6896 | shah.rushabh199@gmail.com | Linkedin | Github

PROFESSIONAL EXPERIENCE

Data Scientist 2 | *Doorstead Inc (Series B Startup), Seattle, USA* **FEB 2022 - FEB 2025**

- **Increased revenue and reduced manual pricing efforts by 70%** by designing and developing **an end-to-end pricing bot** using **ML and reinforcement learning (Q learning)** to dynamically adjust rental property prices based on market conditions and demand.
- Leveraged **LangChain and prompt engineering** to develop an automated pipeline to summarize customer feedback and objections using **LLMS**, and generated weekly reports to provide actionable insights for improving product strategies and customer satisfaction.
- Developed a **predictive model to forecast** rental duration and provide data-driven estimates to customers, resulting in a 5% increase in conversion rates **(measured via A/B testing)** and reducing risk for the company.
- Developed a **RAG chatbot** using LangChain for the owner portal, streamlining onboarding support and reducing escalations by 33%.
- **Reduced model prediction time by 89%** by optimizing the model using **indexing and caching** in Google Vertex AI, which enabled instant offer generation and improved conversion rates by 22%, as measured by **A/B testing**.
- Improved rental pricing **model accuracy by 6%** by developing a **hyperparameter tuning** framework using **Hyperopt**, enabling better optimization of CatBoost algorithms.
- **Reduced cloud costs and runtime by over 15%** by redesigning the ML feature engineering pipeline using **PySpark**.
- **Collaborated** with multiple stakeholders to create and develop **data systems** and **dashboards,** streamlining data ingestion and tracking **KPIs** to identify areas of improvement and enable **A/B experimentation.**

Data Scientist | *CNA Financial (Fortune 500), Chicago, USA* **JUNE 2021 - FEB 2022**

- Developed end-to-end pipeline to **forecast** premiums using **XGboost**, contributing to the company's improved economic structure. **Improved accuracy of prediction by 30%**.
- Collaborated with various teams to develop an **ETL pipeline for the model**, streamlining the process and increasing **efficiency by nearly 20%**.
- **Designed scalable and efficient Python modules** for text matching utilizing cosine similarities and Levenshtein distance, resulting in a 40% improvement in efficiency.

Data Science Intern - Emerging Tech | *Travelers, Hartford, USA* **JUNE 2020 - AUG 2020**

- **Researched, and developed ELECTRA** (**transformer-based LLM**) to automate legal and underwriting data classification. Pretrained the model using legal data and fine-tuned it for the classification task.
- Achieved **97% classification accuracy** and accelerated model training speed by **5X** times by distributing training on **GPUs** reducing training cost.

RELEVANT PROJECTS

Event Recommendation System | *Python, Pytorch, Deep Learning*

- Built a system to recommend personalized events using a pointwise ranking **neural network.**

TECHNICAL SKILLS

Programming Languages | • Python • R • SQL

Libraries/ Frameworks | • Pytorch • Tensorflow • scikit-learn • Pyspark • HuggingFace • Docker • NLTK • MLFLOW

ML Models | • Logistic Regression • SVM • Catboost • Neural Networks • CNN • RNN • BERT

EDUCATION

INDIANA UNIVERSITY BLOOMINGTON, USA **Aug 2019 - May 2021**

Master of Science in Data Science

NMIMS Mumbai, INDIA **July 2015 - June 2019**

Bachelor of Technology in Computer Science